MEDINOVA TEACHING HOSPITAL · CLINICAL CURRICULUM · CLERKSHIP SERIES

# 01

# THE BASELINE ASSESSMENT

*History Taking · Physical Examination · Vital Signs*

*"Before you treat, you must first see."*

ISSUED BY
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TRACK
**Physician & Nurse**
Clerkship Level · Class 01 of 04

# *Structured History Taking*

Every clinical encounter begins with the history. It is not a formality preceding investigation — it **is** the investigation. The history determines which differentials are prioritised, which examinations are targeted, and which investigations are ordered. A clinician who bypasses the history in favour of immediate investigation will order the wrong tests, miss the diagnosis, and expose the patient to unnecessary risk.

## THE SOCRATES FRAMEWORK

APPLIED TO ANY SYMPTOM — PARTICULARLY PAIN, BREATHLESSNESS, AND NEUROLOGICAL PRESENTATIONS

| | COMPONENT | CLINICAL COMMENTARY |
|---|---|---|
| S | **Site** | Where exactly is the symptom? Ask the patient to point with one finger. Localised vs diffuse, unilateral vs bilateral, superficial vs deep. Central crushing chest pain has a fundamentally different differential to left lateral pleuritic chest pain. |
| O | **Onset** | When did it start? Was onset sudden, rapid, or gradual? Sudden-onset severe headache — "the worst of my life" — is subarachnoid haemorrhage until proven otherwise. Sudden-onset central chest pain at rest suggests STEMI or aortic dissection. |
| C | **Character** | What does it feel like? Pressure suggests ischaemia. Burning suggests mucosal irritation or nerve involvement. Tearing or ripping suggests aortic pathology. Colicky pain — building and releasing — suggests hollow viscus obstruction: biliary, renal, or intestinal. |
| R | **Radiation** | Does it move anywhere? Jaw and left arm radiation raises ACS probability significantly. Interscapular radiation with a tearing quality suggests aortic dissection. Radiation to the groin from the flank suggests renal colic. Right shoulder tip pain suggests diaphragmatic irritation. |
| A | **Associated symptoms** | Diaphoresis and nausea with chest pain raise ACS probability substantially. Photophobia and neck stiffness with headache mandate immediate meningitis assessment. Associated symptoms are often more diagnostically discriminating than the presenting symptom itself. |
| T | **Timing** | Constant or intermittent? Worsening, improving, or static? Episodic exertional chest pain relieved by rest is stable angina until proven otherwise. A 3-minute resolved episode of facial droop is a TIA with a 10% stroke risk in the next 48 hours. |
| E | **Exacerbating & relieving factors** | Pleuritic pain — worse on inspiration — suggests PE, pleuritis, or pneumothorax. Relief with GTN suggests ischaemic or spastic aetiology. Pain relieved by sitting forward suggests pericarditis or pancreatitis. |
| S | **Severity** | Rate on a 0–10 numeric scale. Document the number, not a descriptor. Severity alone does not determine urgency — a patient with a large PE may rate their dyspnoea 3/10. The number establishes a baseline for reassessment over time. |

## THE SAMPLE FRAMEWORK — CONTEXTUAL HISTORY

| | COMPONENT | CLINICAL COMMENTARY |
|---|---|---|
| S | **Symptoms** | The presenting complaint in the patient's own words. Document verbatim where possible. "I feel like someone is standing on my chest" is more clinically useful than "chest pain, pressure-like." |
| A | **Allergies** | Drug allergies AND the exact nature of the reaction. "NKDA" must be actively confirmed, not assumed. Distinguish true allergy (immune-mediated) from intolerance — the clinical implications are fundamentally different. |
| M | **Medications** | Full list including dose, frequency, and route. Include OTC medications, herbal supplements, and contraceptive agents. Many serious drug interactions are missed because OTC medications are not asked about. |
| P | **Past medical history** | All previous diagnoses, operations, admissions, and procedures. Previous MI or stroke are independent risk factors for further events. Previous abdominal surgery significantly alters the acute abdomen differential. |
| L | **Last oral intake** | Critical in any patient who may require anaesthesia, procedural sedation, or urgent intervention. Document time and content before the anaesthetist is called. |
| E | **Events leading to presentation** | What were the circumstances immediately before onset? At rest or exerting themselves? Any recent travel, surgery, immobility, or trauma? The context surrounding onset is frequently more informative than the symptom itself. |

CLINICAL PEARL — PROF. C. OBI

*"The history is not a checklist to be completed — it is a conversation to be navigated. The frameworks exist to ensure nothing is missed, not to replace clinical intuition. The most experienced clinicians use SOCRATES automatically, without conscious effort, because they have internalised it entirely. The trainee who reads from a list will miss what the patient is trying to say between the lines. Listen for the pauses. They are often more informative than the words."*

— PROF. CHUKWUEMEKA OBI, CHIEF OF STAFF, MEDINOVA TEACHING HOSPITAL

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# Mental Capacity and Consent

Capacity assessment is a clinical skill, not a legal one, and is among the most commonly mishandled situations in acute medicine. Every clinician who treats adult patients must be able to perform a capacity assessment independently, at the bedside, without legal input.

### THE MENTAL CAPACITY ACT 2005 — TWO-STAGE TEST

| STAGE | QUESTION | CLINICAL APPLICATION |
|---|---|---|
| 1 | **Does this patient have an impairment of, or disturbance in the functioning of, the mind or brain?** | Includes: acute medical conditions (delirium, hypoxia, hypoglycaemia, metabolic disturbance), psychiatric illness, intoxication, and neurodegenerative conditions. This stage asks whether a qualifying condition exists — not whether the patient is incapacitous. Many patients with qualifying conditions retain full capacity. |
| 2 | **Does that impairment cause this patient to be unable to make this specific decision at this time?** | Inability is defined by failure of any one of four functional criteria: (a) Understanding the information relevant to the decision; (b) Retaining that information long enough to use it; (c) Weighing the information as part of the decision-making process; (d) Communicating the decision by any means. Failure of any single criterion constitutes incapacity for that decision. |

### KEY PRINCIPLES

- Capacity is **presumed** — it must be proven absent, not proven present
- Capacity is **time-specific and decision-specific** — it is not a global status
- An unwise decision does not constitute incapacity — adults have the right to make decisions others disagree with
- Next-of-kin cannot consent for an adult — Lasting Power of Attorney for Health and Welfare (LPA HW) is legally required
- Best interests treatment must be the **least restrictive option** that addresses the clinical need
- Reassess capacity regularly as the acute illness resolves — capacity lost to delirium is frequently recoverable

> CLINICAL PEARL — PROF. C. OBI
>
> *"Delirium is reversible. The most common error I see is a clinician documenting 'patient lacks capacity' on admission and never reassessing. Capacity is not a diagnosis — it is a state. A patient with sepsis-induced delirium who lacks capacity on admission may have full capacity 48 hours later once their infection is treated. Reassess. Document each assessment separately with the time, the specific decision being assessed, and which functional criterion is impaired."*
>
> — PROF. CHUKWUEMEKA OBI, CHIEF OF STAFF, MEDINOVA TEACHING HOSPITAL

# Cardiac Auscultation — The Four Valve Areas

Systematic cardiac auscultation follows the four valve areas in a fixed sequence. Use both the diaphragm (high-frequency: S1, S2, most murmurs) and the bell (low-frequency: S3, S4, and the mitral stenosis rumble). Dynamic manoeuvres change murmur intensity and aid diagnosis.

| VALVE | ANATOMICAL POSITION | CLASSIC PATHOLOGY | MURMUR CHARACTER |
|---|---|---|---|
| **Aortic** | 2nd right ICS, right sternal border | Aortic stenosis (AS)<br>Aortic regurgitation (AR) | *AS:* Ejection systolic, radiates to carotids, loudest sitting forward in expiration<br>*AR:* Early diastolic, loudest sitting forward in expiration |
| **Pulmonary** | 2nd left ICS, left sternal border | Pulmonary stenosis<br>Flow murmur (benign) | Ejection systolic, increases with inspiration. Soft systolic murmur in young patients without haemodynamic significance = benign flow murmur |
| **Tricuspid** | Lower left sternal border, 4th–5th ICS | Tricuspid regurgitation (TR) | Pansystolic, increases with inspiration (Carvallo's sign). Distinguishes TR from mitral regurgitation. |
| **Mitral (Apex)** | 5th ICS midclavicular line. Left lateral decubitus. | Mitral regurgitation (MR)<br>Mitral stenosis (MS) | *MR:* Pansystolic, radiates to axilla, diaphragm<br>*MS:* Mid-diastolic rumble with opening snap, bell, left lateral decubitus position |

CLINICAL PEARL — PROF. C. OBI

*"Aortic stenosis is the most common valvular lesion in the Western world. Its classic triad is angina, syncope, and dyspnoea on exertion. Once symptoms develop, median survival without intervention is 2–3 years. Never dismiss a harsh ejection systolic murmur in an elderly patient as age-related sclerosis without formal echocardiographic assessment. The murmur that killed the patient three months later was present at every previous appointment."*

— PROF. CHUKWUEMEKA OBI, CHIEF OF STAFF, MEDINOVA TEACHING HOSPITAL

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# Abdominal Examination — Signs of Peritonism

Peritonism indicates involvement of the parietal peritoneum and constitutes a clinical emergency requiring urgent surgical assessment. It manifests as guarding, rebound tenderness, and board-like rigidity in severe cases. A patient lying very still and reluctant to move has peritonism until proven otherwise.

| SIGN | TECHNIQUE | POSITIVE FINDING | CLINICAL SIGNIFICANCE |
|---|---|---|---|
| Blumberg's sign | Slow deep pressure, then rapid release | Pain on release greater than on application | Parietal peritoneum irritation — appendicitis, peritonitis, any surgical emergency |
| Rovsing's sign | Sustained pressure to left iliac fossa | Pain referred to right iliac fossa | Appendicitis — retrograde pressure transmits to inflamed appendix region |
| McBurney's point | Palpate two-thirds from umbilicus to right ASIS | Maximal tenderness at this point | Appendicitis — point of maximal peritoneal irritation in appendicular inflammation |
| Psoas sign | Left lateral decubitus; extend right hip against resistance | Right iliac fossa or flank pain | Retrocaecal appendix lying on the right psoas muscle |
| Obturator sign | Flex right hip to 90°, internally rotate | Hypogastric or right iliac fossa pain | Pelvic appendix in contact with the obturator internus muscle |
| Murphy's sign | Deep inspiration during right upper quadrant palpation | Inspiratory arrest due to pain | Acute cholecystitis — inflamed gallbladder descends onto examiner's fingers |

# The NEWS2 Scoring System

The National Early Warning Score 2 (NEWS2), published by the Royal College of Physicians (2017), is the validated UK standard for identifying acute clinical deterioration. It is mandated across all NHS trusts and aggregates seven physiological parameters into a composite score that triggers a graded clinical response.

### NEWS2 PARAMETER SCORING TABLE

| PARAMETER | 3 | 2 | 1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| Respiration rate (/min) | ≤8 | – | 9–11 | 12–20 | – | 21–24 | ≥25 |
| SpO2 Scale 1 (%) | ≤91 | 92–93 | 94–95 | ≥96 | – | – | – |
| SpO2 Scale 2 COPD (%) | ≤83 | 84–85 | 86–87 | 88–92 | 93–94 on O2 | 95–96 on O2 | ≥97 on O2 |
| Supplemental O2 | – | Yes | – | No | – | – | – |
| Systolic BP (mmHg) | ≤90 | 91–100 | 101–110 | 111–219 | – | – | ≥220 |
| Heart rate (bpm) | ≤40 | – | 41–50 | 51–90 | 91–110 | 111–130 | ≥131 |
| Consciousness (ACVPU) | – | – | – | Alert | – | – | C/V/P/U * |
| Temperature (C) | ≤35.0 | – | 35.1–36.0 | 36.1–38.0 | 38.1–39.0 | ≥39.1 | – |

* C = New confusion; V = Responds to voice; P = Responds to pain; U = Unresponsive. Any of C/V/P/U scores 3 points.

### CLINICAL RESPONSE THRESHOLDS

| AGGREGATE SCORE | RISK LEVEL | MANDATORY CLINICAL RESPONSE |
|---|---|---|
| 1 – 4 | Low | Minimum 4–6 hourly monitoring. Inform the nurse in charge. Ward-based clinical response. |
| 5 – 6, or any single parameter scoring 3 | Medium | Urgent review by clinician with acute care competency within 30–60 minutes. Consider step-up care. Inform registrar. |
| 7 or above | High | Emergency response. Continuous monitoring. Immediate senior review. Critical care consideration within 30 minutes. Do not delay for investigations. |

CLINICAL PEARL — PROF. C. OBI

*"A NEWS2 score is only as good as the observations that generate it. A respiratory rate counted for 15 seconds and multiplied by four introduces an error of ±4 breaths per minute — enough to shift a NEWS2 score by 2 points in either direction. Count for a full 60 seconds. Post-operative vital sign deterioration is never assumed to be expected. Every abnormal observation is a signal until proven otherwise."*

— PROF. CHUKWUEMEKA OBI, CHIEF OF STAFF, MEDINOVA TEACHING HOSPITAL

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# Sepsis Recognition and Silent Hypoxia

### SEPSIS-3 DEFINITION (SINGER ET AL., JAMA 2016)

Sepsis is defined as **life-threatening organ dysfunction caused by a dysregulated host response to infection**. The previous SIRS criteria have been retired. Organ dysfunction is identified clinically by an acute increase in the Sequential Organ Failure Assessment (SOFA) score of 2 or more points from baseline.

### BEDSIDE SCREENING — QSOFA

Any **two or more** of the following three criteria mandates urgent full SOFA assessment, blood cultures, serum lactate, and immediate escalation:

| QSOFA CRITERION | THRESHOLD |
|---|---|
| Altered mental status | Any new confusion or GCS reduction from baseline |
| Respiratory rate | ≥22 breaths per minute |
| Systolic blood pressure | ≤100 mmHg |

- Lactate **≥2 mmol/L** with clinical sepsis = sepsis requiring resuscitation (Surviving Sepsis Campaign 2021)
- Lactate **≥4 mmol/L** = septic shock regardless of blood pressure — immediate aggressive resuscitation mandatory
- **Silent hypoxia:** SpO2 <94% is a clinical emergency regardless of subjective patient comfort. Objective measurements must never be over-ridden by patient-reported wellbeing

# Pain Assessment Across Patient Populations

Pain is the fifth vital sign and must be assessed with the same rigour as blood pressure. The assessment tool must match the patient's age, cognitive status, and communication ability.

| SCALE | INDICATION | POPULATION | METHOD |
|---|---|---|---|
| Numeric Rating Scale (NRS 0–10) | Self-report | Cognitively intact adults | Patient rates pain 0 (none) to 10 (worst imaginable) |
| Wong-Baker FACES Scale | Self-report with visual aid | Children aged ≥3, mild cognitive impairment | Patient points to the face that matches their pain experience |
| FLACC Scale | Behavioural observation | Infants and children <3, non-verbal children | Score 0–2 each for: Face, Legs, Activity, Cry, Consolability (max 10) |
| Abbey Pain Scale | Behavioural observation | Non-verbal adults with severe dementia | Score 6 domains: vocalisation, facial expression, body language, physiological change, physical change, behaviour change |

# Paediatric Assessment — Unique History Components

Children are not small adults. The paediatric history contains components with no adult equivalent, and paediatric vital sign normal ranges differ substantially from adult reference values at every age group.

### UNIQUE COMPONENTS

- **Birth history:** gestation, birth complications, NICU admission, birth weight. Prematurity affects baseline vital signs and respiratory vulnerability throughout childhood
- **Developmental milestones:** is the child meeting age-appropriate targets for motor, language, social, and cognitive function? Developmental regression is always significant
- **Immunisation history:** determines which vaccine-preventable diseases remain possible. An unimmunised febrile child cannot have meningococcal disease excluded without active investigation
- **Feeding history:** in infants, reduced feeding is a sensitive early indicator of illness severity. Three consecutive refused feeds in a breastfed infant is clinically significant

### NON-BLANCHING RASH — RED FLAG

Any non-blanching rash in a febrile child is **meningococcal septicaemia until proven otherwise**. This is a time-critical medical emergency.

**Tumbler test:** press a clear glass firmly against the skin. A rash visible through the glass under pressure = non-blanching. Do not wait for the rash to reappear if the history describes one.

**Immediate action:** blood cultures, IV access, empirical IV ceftriaxone 50 mg/kg (max 2 g). Do not delay antibiotics for imaging or further investigation.

> CLINICAL PEARL — PROF. C. OBI
>
> *"Unrecognised pain in patients with dementia is one of the most prevalent and entirely preventable causes of agitated behaviour on a hospital ward. A patient who cannot tell you they are in pain is not a patient without pain — they are a patient who cannot communicate it. Behavioural disturbance in dementia must always prompt formal pain assessment before any sedative intervention is considered. The Abbey score takes ninety seconds to complete. There is no excuse for omitting it."*
>
> — PROF. CHUKWUEMEKA OBI, CHIEF OF STAFF, MEDINOVA TEACHING HOSPITAL

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# *Emergency Triage — Manchester Triage System*

The Manchester Triage System (MTS) allocates every patient to one of five priority categories based on presenting complaint and discriminator assessment. Triage is dynamic — a patient's category must be formally reassessed if their condition changes while waiting.

| CATEGORY | COLOUR | MAX TIME | CLINICAL EXAMPLES |
|---|---|---|---|
| Immediate | Red | 0 min | Cardiac arrest, airway obstruction, major haemorrhage, non-blanching rash in febrile child, GCS <8 |
| Very Urgent | Orange | 10 min | Chest pain with haemodynamic compromise, severe breathlessness (SpO2 <90%), altered consciousness, high-risk injury mechanism |
| Urgent | Yellow | 60 min | Moderate acute pain, significant injury without haemodynamic instability, high-risk history without current compromise |
| Standard | Green | 120 min | Minor injury, normal vital signs, low-risk history, chronic stable presentations |
| Non-Urgent | Blue | 240 min | Prescription requests, chronic administrative presentations, minor complaints of very recent onset |

Under-triage is the most dangerous triage error. Under-triage rates above 5% are considered unacceptable. When in doubt, assign the higher category. Over-triage delays less unwell patients; under-triage kills.

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